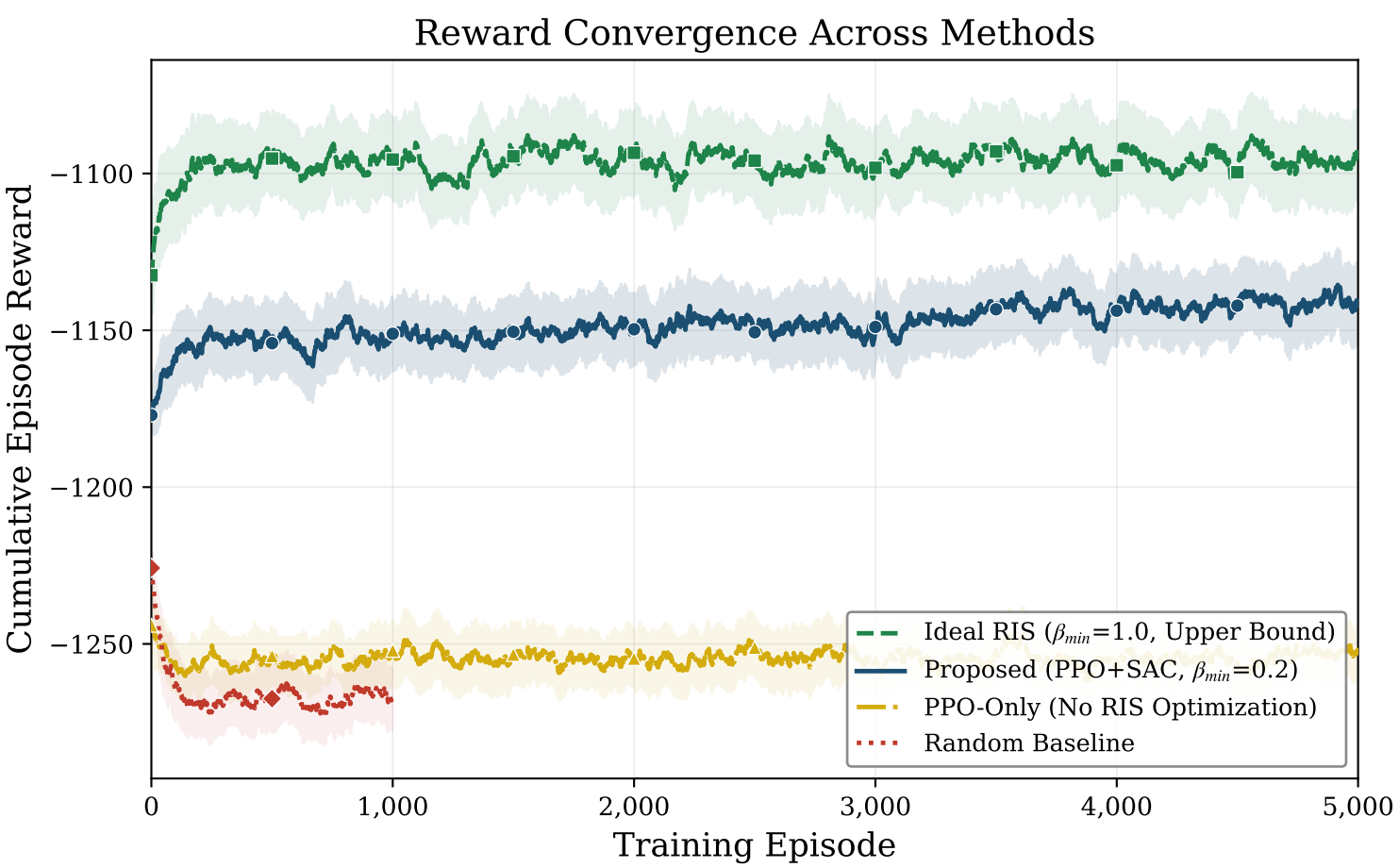

Reward Convergence Across Methods
Cumulative Episode Reward
−1100
−1150
−1200
−1250
0
1,000
2,000
3,000
4,000
5,000
Training Episode
Ideal RIS (β_{min}=1.0, Upper Bound)
Proposed (PPO+SAC, β_{min}=0.2)
PPO-Only (No RIS Optimization)
Random Baseline